

SOMBOR, SerbiaWMO: **131600**Lat: **45.767N**Lon: **19.150E**Elev: **89**StdP: **100.26**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-8.0	-13.8	1.1	-8.6	-11.1	1.5	-6.2	8.3	2.2	7.5	1.7	1.6	0	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	33.8	20.6	32.2	20.0	30.6	19.5	22.2	29.4	21.4	28.6	20.7	27.7	2.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	15.2	23.8	19.4	14.3	23.2	18.5	13.5	22.7	65.3	29.0	62.6	28.7	60.0	27.6	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.4	5.3	-16.1	36.1	5.3	1.8	-19.9	37.4	-23.0	38.5	-26.0	39.5	-29.9	40.9		
(5)			-16.3	23.5	5.4	1.1	-20.1	24.2	-23.3	24.9	-26.3	25.5	-30.3	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	0.7	2.4	7.0	12.5	17.1	20.6	22.5	22.3	17.4	12.1	7.0	1.6
	DBStd	8.77	4.64	5.26	4.53	3.79	3.68	3.45	2.99	3.12	3.52	4.25	4.31	4.50
	HDD10.0	1037	289	216	113	21	1	0	0	0	0	28	109	261
	HDD18.3	2728	547	447	352	178	67	16	4	5	59	195	339	518
	CDD10.0	1763	1	2	19	94	222	319	389	382	222	93	19	1
	CDD18.3	414	0	0	0	2	29	86	134	129	31	3	0	0
	CDH23.3	4458	0	0	1	40	325	908	1469	1358	321	37	0	0
	CDH26.7	1723	0	0	0	6	80	327	631	587	92	2	0	0
Wind	WSAvg	2.3	2.3	2.5	2.8	2.7	2.4	2.1	2.1	1.8	1.9	2.0	2.2	2.3
Precipitation	PrecAvg	632	36	37	34	42	67	74	68	55	63	56	52	48
	PrecMax	979	76	88	100	106	221	209	161	145	143	132	130	105
	PrecMin	290	4	2	3	0	19	12	20	6	16	5	3	1
	PrecStd	142	20	22	22	23	43	44	34	31	37	36	27	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	17.5	22.0	27.4	31.2	34.1	35.8	36.5	31.7	26.5	20.5	13.7
		MCWB	9.7	9.8	12.1	15.9	18.2	21.4	21.4	21.1	18.7	16.5	13.5	9.9
	2%	DB	10.7	14.4	19.3	24.0	28.9	32.3	33.7	34.0	29.3	24.2	17.7	11.0
		MCWB	7.8	8.6	10.9	13.8	17.6	20.6	20.6	20.4	17.8	15.4	12.6	8.7
	5%	DB	8.5	11.8	16.9	22.0	27.0	30.4	32.1	32.0	27.1	22.2	15.5	9.0
		MCWB	6.4	7.7	9.9	12.8	16.7	19.9	20.3	19.6	17.0	14.6	11.7	7.2
	10%	DB	6.7	9.4	14.5	20.1	24.9	28.6	30.3	30.1	24.8	19.8	13.5	7.3
		MCWB	5.3	6.6	8.8	12.1	15.9	19.0	19.5	19.2	16.2	13.8	10.7	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	10.9	12.8	16.5	19.7	23.2	23.0	23.0	20.0	17.9	14.7	10.5
		MCDB	12.6	14.9	19.4	24.8	26.7	30.0	30.3	30.3	26.8	23.7	18.3	12.8
	2%	WB	8.2	9.3	11.7	15.0	18.7	21.9	22.1	21.9	19.1	16.6	13.3	8.8
		MCDB	10.3	13.2	17.7	21.3	25.7	28.7	29.2	29.6	25.7	22.0	16.4	10.9
	5%	WB	6.6	8.1	10.5	13.9	17.9	21.0	21.5	21.1	18.2	15.5	12.1	7.5
		MCDB	8.1	11.0	15.5	19.7	24.4	27.7	28.6	28.5	23.9	20.3	15.0	8.8
	10%	WB	5.4	7.0	9.5	12.9	17.0	20.1	20.7	20.3	17.4	14.4	10.8	6.1
		MCDB	6.7	9.2	13.8	18.3	22.8	26.7	27.7	27.2	22.6	18.6	13.3	7.2
Mean Daily Temperature Range	5% DB	MDBR	5.9	7.8	9.9	10.9	11.1	11.4	12.1	12.7	11.4	10.7	8.0	5.5
		MCDBR	9.3	12.4	14.2	15.1	14.9	14.7	15.6	16.2	15.5	14.4	11.7	8.4
		MCWBR	6.3	7.2	7.0	6.3	5.3	4.7	4.5	4.8	5.2	6.5	6.6	6.0
	5% WB	MCDBR	8.7	10.5	12.3	12.8	12.8	12.8	13.3	13.8	12.6	12.7	10.5	7.9
		MCWBR	6.2	6.7	6.5	5.8	5.1	4.7	4.5	4.9	4.9	6.3	6.4	5.9
Clear-Sky Solar Irradiance	taub		0.340	0.364	0.398	0.453	0.440	0.437	0.443	0.447	0.429	0.394	0.361	0.340
	taud		2.361	2.289	2.229	2.081	2.183	2.227	2.203	2.198	2.218	2.310	2.357	2.366
	Ebn at Noon		748	798	822	805	831	834	822	802	780	753	712	702
	Edh at Noon		77	99	121	153	142	137	139	134	121	96	77	69
All-Sky Solar Radiation	RadAvg		1.19	2.01	3.25	4.62	5.61	6.13	6.16	5.50	3.83	2.60	1.49	0.95
	RadStd		0.18	0.29	0.41	0.43	0.49	0.49	0.46	0.54	0.43	0.27	0.14	0.14